

# SUDHARSAN V *Software Engineer*

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## PROFILE

Full-Stack Software Engineer with hands-on experience building, scaling, and deploying production web applications using the MERN stack and Next.js. Experienced in developing secure REST APIs, real-time architectures with sub-60ms socket latency, and high-performance interfaces scoring 98+ on Lighthouse.

## EDUCATION

**PSN College of Engineering and Technology**  
*Bachelor of Engineering in Computer Science and Engineering*

Tirunelveli, Tamil Nadu  
Sep 2023 – May 2027

## TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), HTML5, CSS3, SQL
- **Frontend:** React.js, Next.js, Tailwind CSS, TanStack Query, Responsive UI, Web Performance Optimization
- **Backend:** Node.js, Express.js, RESTful APIs, JWT Auth, Role-Based Access Control (RBAC), Rate Limiting, API Security
- **Databases:** MongoDB, Mongoose, Database Schema Design, Query Optimization
- **Real-Time & Integrations:** Socket.IO, ZegoCloud, Razorpay, Firebase, Cloudinary, SMTP, MSG91
- **Tools & Deployment:** Git, GitHub, Docker, Postman, Linux, Vercel, Render, Hostinger VPS
- **Core Concepts:** Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems, Computer Networks

## EXPERIENCE

**Tetron Infotech**  
*Full Stack Developer Intern*

Sivagangai, Tamil Nadu  
April 2024 – May 2024

- **Real-Time Architecture:** Architected bidirectional communication channels using **Socket.IO** and **Node.js**, delivering real-time presence detection and message delivery for **500+ concurrent connections** with **sub-60ms latency**.
- **State Synchronization:** Developed modular chat interfaces in **Next.js** and **React**, implementing **TanStack Query** for client-side state caching to reduce redundant network API requests by **40%**.
- **API Architecture & Security:** Engineered **12+ secure RESTful endpoints** with **Firebase Admin Auth**, securing token validation for **1,000+ sessions** while integrating **ZegoCloud** and cutting deployment overhead by **30%** via automated pipelines.

## PROJECTS

**DevTackle** | [Live Demo](#) | [GitHub](#) | *React, Node.js, Express, MongoDB, Socket.IO, Razorpay* 2024

- Architected a scalable full-stack community platform for **500+ developers**, integrating **Socket.IO** to broadcast real-time discussions, voting, and activity notifications with sub-60ms delivery.
- Integrated **Razorpay** payment gateway with server-side webhook signature verification; implemented dual-token JWT authentication (**15-min access + 7-day refresh token rotation**) with rate limiting.
- Optimized MongoDB schemas with compound indexing to speed up search queries by **45%**, achieving a **98/100 Lighthouse performance score** and **< 1.1s LCP** via dynamic code-splitting.

**Bhairava Jobs** | [Live Demo](#) | [GitHub](#) | *Next.js, Node.js, Express, MongoDB, Cloudinary* 2024

- Architected a production full-stack commercial job portal connecting **20+ local businesses** with **500+ active job seekers**, reducing listing-to-application latency by **35%** through dynamic categorization.
- Hardened backend REST APIs with **100 req/15-min rate limiting**, **Helmet** HTTP security headers, and **Joi** validation, preventing NoSQL injection and maintaining **sub-80ms API latency**.
- Streamlined media upload pipelines using **Multer** and **Cloudinary** streaming (reducing server disk overhead to **0MB**) and implemented structured request tracking with **Pino** logger.

## OPEN SOURCE

- **DevBoard** | *Merged Pull Request (#346)*: Hardened backend API security by refactoring CORS policy across **15+ routes**, eliminating cross-origin vulnerabilities for **1,000+ platform users** while preserving credentialed requests.